

Siddhant Agarwal

Bangalore, India — siddhantagarwal0401@gmail.com

EDUCATION

Stonehill International School)

IBDP

- 75% merit-based scholarship recipient.

Aug 2025 – Present

Bangalore, India

Vibgyor High School

ICSE

- GPA: 4.0 — Class Rank: 4 / 350.

Jun 2011 – Mar 2025

Bangalore, India

RELEVANT EXPERIENCE

Independent Research Project — Comparative Study of CNNs and ViTs

Aug 2025 – Present

Extended Essay

- Designing controlled experiments to compare convolutional neural networks and vision transformers for image classification.
- Evaluating model robustness under increasing levels of Gaussian blur to analyse performance degradation patterns.
- Analysing accuracy trends and architectural differences to draw insights on inductive bias and generalisation behaviour.

Founder & Program Lead

Aug 2025 – Present

ECCP Robotics Program (FIRST Tech Challenge)

Bangalore, India

- Founded and led a robotics program to train students for participation in the FIRST Tech Challenge.
- Designed training modules covering robotics, programming logic, and iterative engineering design.

Software Development Intern

June 2025 – Present

eVidyaLoka

Bangalore, India

- Built AI-driven automation workflows to streamline internal processes and reduce manual effort.
- Worked on frontend development for educational platforms with a focus on usability and accessibility.
- Collaborated with cross-functional teams to deploy features used in rural education settings.

AI-Driven Full-Stack Project

May 2025 – Jan 2026

Founder: MediScan

- Designed an end-to-end system to digitise handwritten medical prescriptions using OCR and AI-based text processing.
- Extracted structured medication data and generated simplified, patient-friendly explanations.
- Integrated automation workflows to support accessibility features such as audio-based explanations.

Computer Science Teaching Intern

June 2025 – Jul 2025

Vidya

Bangalore, India

- Taught foundational computer science concepts, including algorithms, logic, and problem-solving.
- Designed interactive lessons to communicate technical ideas clearly to non-technical audiences.

Student Research Project – Solar Energy

Aug 2025 – Dec 2025

Junior Academy (New York Academy of Sciences)

- Collaborated in a global interdisciplinary team to design a solar-based solution to real-world energy challenges.
- Applied physics principles, data analysis, and modelling to evaluate system feasibility.

Delegate & Award Recipient

Nov 2022 – Jan 2025

Model United Nations (MUN)

- Researched geopolitical, economic, and technological policy issues.
- Drafted structured position papers under strict time constraints.
- Won 3 awards in 5 conferences.

AWARDS & HONORS

Bronze Medalist Infinity Math Competition	2026
Connect Award Winner FIRST Tech Challenge (FTC)	2026
Finalist Imperial STEMathon	2025
First Place Meraki Hackathon	2025
Global Indian Champion Harvard Science Bowl	2025
State-Level Player Basketball	2023-2025

MEDIA & PUBLICATIONS

Technical Writing on AI & Computer Science Medium	Sep 2024 – Present
• Publication overview: Authored 20 articles with 3000+ cumulative views on artificial intelligence, machine learning, and computer vision.	
Selected Articles	
• <u>The AI Wall: Examining Today’s Limitations</u> — analysed current practical and theoretical constraints in modern AI systems. • <u>LLMs Explained: How AI Like ChatGPT Understands Language</u> — explained the mechanisms by which large language models process and generate human-like text. • <u>Model Context Protocol (MCP) Explained Simply</u> — explained the Model Context Protocol and its role in enhancing long-term memory and efficiency in AI systems.	
Long-Form Essays on Technology & Society Substack	Oct 2024 – Present
• Wrote reflective essays examining the societal and ethical implications of emerging technologies. • <u>Is ChatGPT Making You Dumb? The Truth</u> — critically examined cognitive effects and learning behavior in users interacting with large language models.	

TECHNICAL SKILLS

Programming Languages	Python, JavaScript, TypeScript, SQL
Frameworks & Libraries	Django, Express.js, React Native, Expo
AI & Data	OCR Pipelines, AI Automation Workflows, Prompt Engineering, Basic Machine Learning
Robotics & Systems	FTC Robotics, Iterative Engineering Design, System Integration
Tools & Platforms	Git, Linux, Supabase, Firebase